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| <b>B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19</b><br>(Autonomous Institute, Affiliated to VTU)<br><b>Computer Science and Engineering</b> |                                           |                        |
| <b>SECOND INTERNALS</b>                                                                                                                    |                                           |                        |
| <b>Course Code: 23CS4PCSED</b>                                                                                                             | <b>Course Title: Software Engineering</b> |                        |
| <b>Semester: 4<sup>th</sup> A,B,C,D,E,F</b>                                                                                                | <b>Maximum Marks: 40</b>                  | <b>Date: 29/6/2024</b> |
| <b>Faculty Handling the Course:</b>                                                                                                        | <b>Dr.LNR, SWT, RM, SB, SAK, RKM</b>      |                        |

**Instructions:** i) *Answer all Part A and Part B Questions*  
 ii) *In Part C - Answer 3a or 3b and 4a or 4b*

**PART –A**

| No. | Question | Marks | CO | PO |
|-----|----------|-------|----|----|
|-----|----------|-------|----|----|

**1** Explain the organizational paradigms as suggested by Constantine for defining the structure of the software team.

not perform as efficiently as other teams.  
 4. A synchronous paradigm relies on the natural compartmentalization of a problem and organizes team members to work on pieces of the problem with little active communication among themselves.

**M**

**SOLUTION:**

1. A closed paradigm structures a team along a traditional hierarchy of authority. Such teams can work well when producing software that is quite similar to past efforts, but they will be less likely to be innovative when working within the closed paradigm

2. A random paradigm structures a team loosely and depends on individual initiative of the team members. When innovation or technological breakthrough is required, teams following the random paradigm will excel. But such teams may struggle when “orderly performance” is required.

3. An open paradigm attempts to structure a team in a manner that achieves some of the controls associated with the closed paradigm but also much of the innovation that occurs when using the random paradigm. Work is performed collaboratively, with heavy communication and consensus-based decision making the trademarks of open paradigm teams. Open paradigm team structures are well suited to the solution of complex problems but may

**PART –B**

| No. | Question                                                                                                                             | M<br>a<br>r<br>k<br>s |   | PO |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---|----|
| 2a  | Differentiate between Enduring and Volatile requirements. Identify enduring and volatile requirements for Hospital management system | 5                     | 2 | 2  |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|  | <p><b>SOLUTION</b></p> <p>1. Enduring requirements These are relatively stable requirements that derive from the core activity of the organization and which relate directly to the domain of the system. For example, in a hospital, there will always be requirements concerned with patients, doctors, nurses and treatments. These requirements may be derived from domain models that show the entities and relations that characterize an application domain</p> <p>2. Volatile requirements These are requirements that are likely to change during the system development process or after the system has been become operational. An example would be requirements resulting from government healthcare policies.</p> <p><b>Enduring Requirements:</b></p> <ol style="list-style-type: none"> <li><b>Security and Privacy:</b> <ul style="list-style-type: none"> <li><b>Enduring:</b> Ensuring patient data confidentiality, integrity, and availability.</li> <li><b>Importance:</b> Critical to comply with laws and ethical standards, protecting patient information from unauthorized access.</li> </ul> </li> <li><b>Patient Management:</b> <ul style="list-style-type: none"> <li><b>Enduring:</b> Core functionalities such as patient registration, admission, discharge, and transfer (RADT).</li> <li><b>Importance:</b> Essential for managing patient care processes and maintaining accurate medical records.</li> </ul> </li> <li><b>Billing and Financial Management:</b> <ul style="list-style-type: none"> <li><b>Enduring:</b> Features for billing, insurance claims processing, and financial reporting.</li> <li><b>Importance:</b> Integral for revenue management and financial transparency within the healthcare organization.</li> </ul> </li> </ol> <p><b>Volatile Requirements:</b></p> <ol style="list-style-type: none"> <li><b>Technology and Integration:</b> <ul style="list-style-type: none"> <li><b>Volatile:</b> Evolving hardware and software technologies, integration with new healthcare IT systems.</li> <li><b>Importance:</b> Constant adaptation to leverage new technological advancements for enhanced efficiency and patient care.</li> </ul> </li> <li><b>Regulatory Compliance:</b> <ul style="list-style-type: none"> <li><b>Volatile:</b> Changes in healthcare regulations (e.g., HIPAA, GDPR) and compliance requirements.</li> </ul> </li> </ol> |  |  |  |
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|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |   |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|
|    | <ul style="list-style-type: none"> <li>○ <b>Importance:</b> Continuous updates to ensure adherence to legal standards and avoid penalties.</li> </ul>                                                                                                                                                                                                                                                                                                     |        |   |   |
| 2b | <p>Analyze why design conflicts might arise when designing an architecture where availability and security requirements are the most important functional requirements. Justify your answer.</p> <p><b>SOLUTION</b></p> <p>If availability is the critical requirement and is to be included in design then the subsystems should be redundantly placed in many places so that if one system fails another will take over and will still be available</p> | 5<br>M | 2 | 2 |

|  |                                                                                                                                                                                                   |  |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|  | <p>If security is the critical requirement, then the security aspects should be put in one subsystem only. Thus, conflicts arise when both are to be included, and some compromise to be done</p> |  |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|

2c Analyze the series of questions to be answered that lead to a definition of key project characteristics and the resultant project plan.

5

M

**SOLUTION**

- Why is the system being developed? All stakeholders should assess the validity of business reasons for the software work. Does the business purpose justify the expenditure of people, time, and money?
- What will be done? The task set required for the project is defined.
- When will it be done? The team establishes a project schedule by identifying when project tasks are to be conducted and when milestones are to be reached. Who is responsible for a function? The role and responsibility of each member of the software team is defined.
- Where are they located organizationally? Not all roles and responsibilities reside within software practitioners. The customer, users, and other stakeholders also have responsibilities
- How will the job be done technically and managerially? Once product scope is established, a management and technical strategy for the project must be defined.

How much of each resource is needed? The answer to this question is derived by developing estimates.

**PART –C**  
2 2

No. Question Mark

CO

s

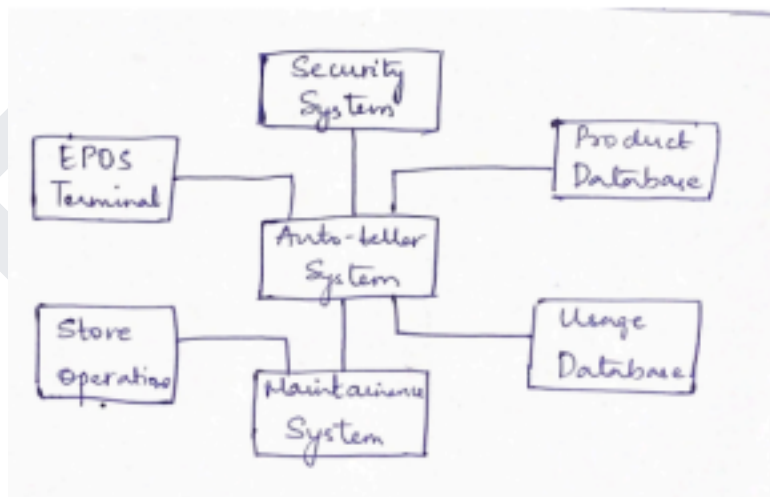
3a (i) Assume you are responsible for writing the specification for a software system that controls a network of EPOS (electronic point of sale) terminals in a store. The system accepts bar code information from a terminal, queries a product database and returns the item name and its price to the terminal for display. The system must be continually available during the store's

opening hours. Draw Context model and Dataflow model

SOLUTION

CONTEXT MODEL

10 3



Data flow Model  
Carries 2.5 marks

(ii) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate structural model (How a system is decomposed into sub-systems) that might be used as a basis for managing the development of the following systems: 1) An

University Automated Marks card Generation system

2) An automated robot floor cleaner

Explain with neat diagram

SOLUTION

1) An University Automated Marks card Generation system

Centralized Repository/ Client-server model

2) An automated robot floor cleaner

The most appropriate model is a repository model where all sub-systems place information in the repository for other sub-systems to use. In the case of AI systems as this would be, a special kind of repository called a blackboard is normally used.

OR

3b (i) Giving reasons for your answer suggest an appropriated control model for the following with a neat diagram:

1) A system that monitors sugar level in patients and dispenses insulin

2) A system to control a Television unit

Explain with neat diagram

SOLUTION

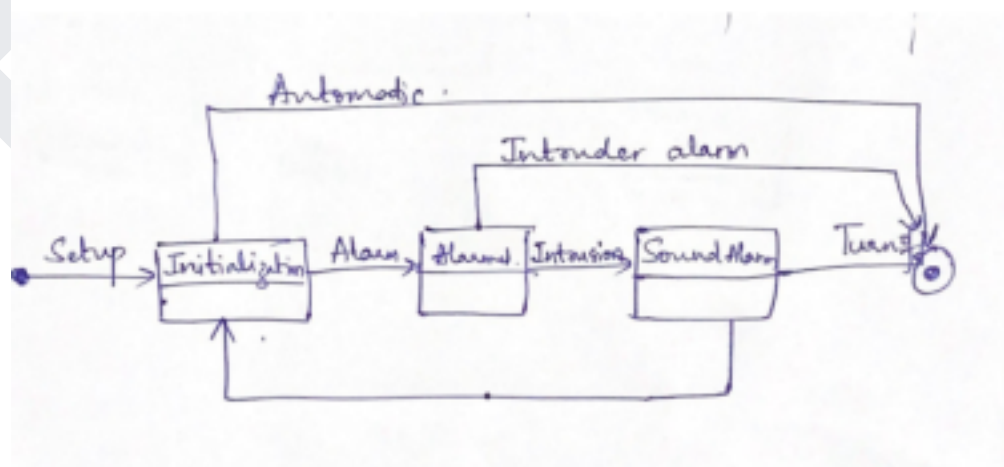
1) A system that monitors sugar level in patients and dispenses insulin

Manager Model

2) A system to control a Television unit

Centralized (polling) control model. This is the most appropriate approach as there is no need for the very rapid response required from interrupt driven systems.

(ii) Draw a state machine model for an automatic intruder alarm and lighting control system.



4a (i) An organizations average productivity is 1250 LOC/pm. The average labor rate is \$12250 per month. Calculate

- 1) Cost per line of Code
- 2) Overall Project cost if the proposed project has 276000LOC
- 3) Estimated effort in person-months

#### SOLUTION

i) To calculate the cost per line of code (LOC),

Cost per LOC = Labor rate / Average productivity

Cost per LOC = \$12250 / 1250LOC/pm

Cost per LOC  $\approx$  \$9.8 per LOC

ii) To calculate the overall project cost,

Overall project cost = Proposed project LOC \* Cost per LOC

Overall project cost = 276000 LOC \* \$9.8 per LOC

Overall project cost  $\approx$  \$27,04,800

iii) To estimate the effort in person-months

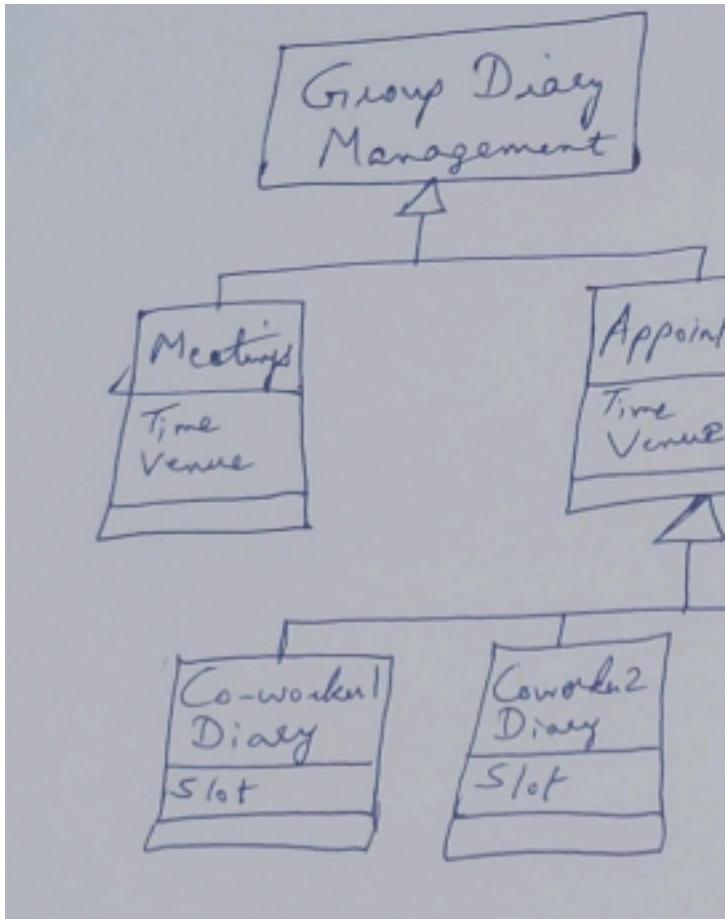
10 3 10 3

Estimated effort in person-months = Proposed project LOC /  
Average productivity Estimated effort in person-months =  
276000LOC / 1250LOC/pm  
=221 person-months.

(ii) Using the UML graphical notation for object classes,  
Identify the objects and design object Generalization diagram  
showing interactions of objects for the following:  
A group diary and time management system is intended to

support the timetabling of meetings and appointments across a group of coworkers. When an appointment is to be made that involves a number of people, the system finds a common slot in each of their diaries and arranges the appointment for that time. If no common slots are available it interacts with the user to rearrange his or her personal diary to make room for the appointment.

#### SOLUTION



Note: Also interactions to be written(drawn)

#### OR

4b (i) If an organization's productivity is 12 FP/pm based on a burdened labor rate of \$20,000 per month. The Count total 450 and  $\sum (f_i)$  is 58. Calculate:

- 1) Cost per FP
- 2) Estimated effort in person-months
- 3) Estimated project cost

#### SOLUTION

10 3

FP estimate=count total\*[0.65+0.01\*£(Fi)]

= 450\*[0.65+0.01\*58]

=450\*1.23

= 553.5 FP

Total FP estimate= 554 FP

1)Cost per Function point = Labour rate per month /Organization Avg Productivity  
= \$20000 per month/ 12FP

=1667 FP

2)Effort in Person Month= Total FP/ Organization Avg Productivity

$$=554 \text{ FP}/12$$

$$=46$$

3) Estimated project cost = (Total FP) \* (cost per FP)

$$= 554 * 1667$$

$$=\$9,23,518$$

(ii) Draw sequence diagram that shows possible sequence of actions that occur when a new article is catalogued by the LIBSYS (Library management system).

### SOLUTION

The sequence diagram in is concerned with the issue of electronic items where a LIBSYS user looks up items in a catalogue then has them delivered to their computer. A licence for use is issued before the items can be delivered. Tests that might be developed are:

1. Lookup the catalogue for an item that you know is present.

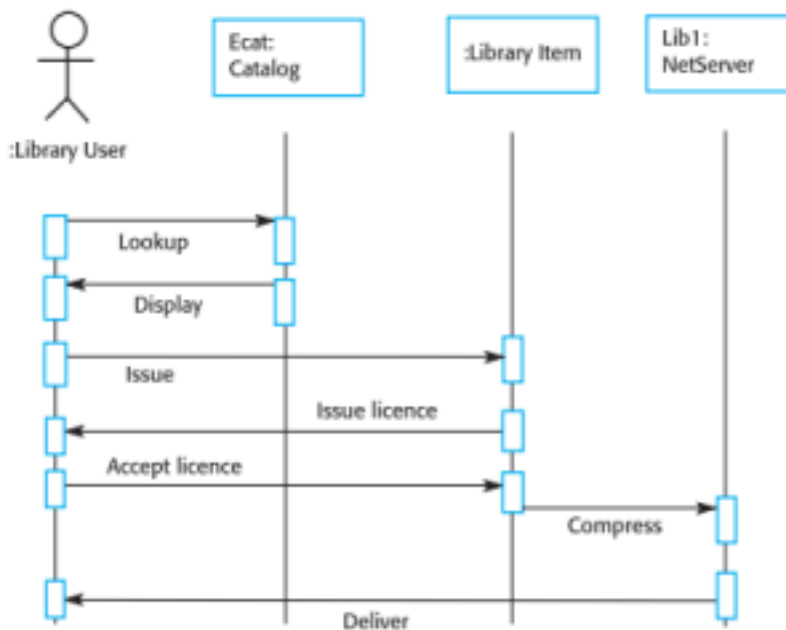
2. Lookup the catalogue for an item that you know is not present.

3. Ask for a catalogue item to be issued. Accept the licence

4. Ask for a catalogue item to be issued. Refuse the licence.

5. Test that a compressed item has been downloaded.

6. Test that a compressed item has been properly decompressed by the computer receiving the item.



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|  | <div>BMSCOLLEGE OF ENGINEERING, BANGALORE-19</div> <div>(Autonomous Institute, Affiliated to VTU)</div> <div>Department Name:</div> |                  |  |
| SECOND INTERNALS                                                                    |                                                                                                                                     |                  |  |
| Course Code: 20CS5PCSEG                                                             | Course Title: Software Engineering                                                                                                  |                  |  |
| Semester: 5 <sup>th</sup> A, B, C                                                   | Maximum Marks: 40                                                                                                                   | Date: 01/12/2020 |  |
| Faculty Handling the Course: Latha N.R., Sheetal V.A.                               |                                                                                                                                     |                  |  |

**PART-A**

**Total 5 Marks**

| No. | Question                                                          | Marks |
|-----|-------------------------------------------------------------------|-------|
| 1   | Describe the strengths and weaknesses of using structured methods | 5M    |

**PART-B**

**Total 15 Marks**

| No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2a  | Using the UML graphical notation for object classes, design state diagram showing the possible state changes in one or more of the objects that you have defined for the following :<br><br>(i) A vending machine that can dispense coffee with and without milk and sugar. The user deposits a coin and makes his or her selection by pressing a button on the machine. This causes a cup with powdered coffee to be output. The user places this cup under a tap presses another button and hot water is dispensed. | 5M    |
| 2b  | Differentiate between Incremental development over prototyping.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5M    |
| 2c  | Analyze why design conflicts might arise when designing an architecture where availability and security requirements are the most important functional requirements. Justify your answer.                                                                                                                                                                                                                                                                                                                             | 5M    |

**PART- C**

**Total 20 marks**

| No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3a  | A safety-critical software system for treating cancer patients has two principal components:<br><br><ul style="list-style-type: none"> <li>• A radiation therapy machine that delivers controlled doses of radiation to tumor sites, this machine is controlled by an embedded software system</li> <li>• A treatment database that includes details of the treatment given to each patient. Treatment requirements are entered in this database and are automatically downloaded to the radiation therapy machine</li> </ul><br>(i) Model architecture for the above system with reasonable assumptions about the system requirements and justify your answer. | 10M   |

|           |                                                                                                        |     |
|-----------|--------------------------------------------------------------------------------------------------------|-----|
|           | (ii) Draw Context Model and State Machine model for the above system                                   |     |
| OR        |                                                                                                        |     |
| <b>3b</b> | (i) Draw a sequence model, data model and state machine model for an automatic Coffee vending machine. | 10M |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>4a</b> | <p>(i) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate structural model that might be used as a basis for managing the development of the following systems:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> An Automated Marks card Generation system</li> <li><input type="checkbox"/> An automated robot floor cleaner</li> </ul> <p>Explain with neat diagram.</p> <p>(ii) Suggest the most appropriate control model that might be used for the following systems. Provide proper justification for your answers. <input type="checkbox"/> An automated car driving system</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> A library catalogue system that prints requested articles</li> </ul> <p>Explain with neat diagram.</p>                                                                                        | 10M |
| OR        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
| <b>4b</b> | <p>(ii) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate structural model that might be used as a basis for managing the development of the following systems:</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> An interactive system that allows railway passengers to find train times from terminals installed in stations</li> <li><input type="checkbox"/> A university system that prints marks cards</li> </ul> <p>Explain with neat diagram.</p> <p>(ii) Suggest the most appropriate control model that might be used for the following systems. Provide proper justification for your answers. Explain with neat diagram.</p> <ul style="list-style-type: none"> <li><input type="checkbox"/> An Automated radiation therapy system</li> <li><input type="checkbox"/> Digital Library system</li> </ul> <p>Explain with neat diagram.</p> | 10M |

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|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------|
|  | <div>BMSCOLLEGE OF ENGINEERING, BANGALORE-19</div> <div>(Autonomous Institute, Affiliated to VTU)</div> <div>Department Name:</div> |                                    |                  |
| SECOND INTERNALS                                                                    |                                                                                                                                     |                                    |                  |
| Course Code: 20CS5PCSEG                                                             |                                                                                                                                     | Course Title: Software Engineering |                  |
| Semester: 5 <sup>th</sup> A,B,C                                                     |                                                                                                                                     | Maximum Marks: 40                  | Date: 01/12/2020 |

Faculty Handling the Course: Latha N.R., Sheetal V.A.

Instructions: Internal Choice in PART C only

### PART-A

Total 5 Marks

| No. | Question                                                                                      | Marks |
|-----|-----------------------------------------------------------------------------------------------|-------|
| 1   | Describe the strategies used for decomposing a sub-system into modules with relevant diagrams | 5M    |

### PART-B

Total 15 Marks

| No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2a  | Using the UML graphical notation for object classes, design sequence diagram showing interactions of objects for the following :<br><br>(i) A group diary and time management system is intended to support the timetabling of meetings and appointments across a group of coworkers. When an appointment is to be made that involves a number of people, the system finds a common slot in each of their diaries and arranges the appointment for that time. If no common slots are available it interacts with the user to rearrange his or her personal diary to make room for the appointment. | 5M    |
| 2b  | Differentiate between the two kinds of concurrent objects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5M    |
| 2c  | Analyze the various schemes that help in identifying objects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5M    |

### PART- C

Total 20 marks

| No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3a  | Your customer wants to develop a system for stock information where dealers can access information about companies and can evaluate various investment scenarios using a simulation system. Each dealer uses this simulation in a different way, according to his or her experience and the type of stocks in question.<br><br>(i) Model architecture for the above system with reasonable assumptions about the system requirements and justify your answer.<br>(ii) Draw Context model and Sequence model | 10M   |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| OR        |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>3b</b> | <p>(i) Draw a Data flow model showing the interactions involved when a student's registers for a course in a university. Courses may have limited enrolment, so that the registration process must include checks that places are available. Assume that the student accesses an electronic course catalogue to find out about various courses</p> <p>(ii) Draw a state machine model for an automatic intruder alarm and lighting control system.</p> | 10M |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>4a</b> | <p>(i) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate structural model that might be used as a basis for managing the development of the following systems:</p> <ul style="list-style-type: none"> <li>□ Passport Authentication system</li> <li>□ A system to control Air traffic controller</li> </ul> <p>Explain with neat diagram.</p> <p>(ii) Suggest the most appropriate control model that might be used for the following systems. Provide proper justification for your answers. □ An interactive system that allows railway passengers to find train times from terminals installed in stations</p> <ul style="list-style-type: none"> <li>□ A university system that prints marks cards</li> </ul> <p>Explain with neat diagram.</p> | 10M |
| □ OR      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |
| <b>4b</b> | <p>(ii) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate structural model that might be used as a basis for managing the development of the following systems:</p> <ul style="list-style-type: none"> <li>□ A system that monitors patients in a hospital intensive care unit</li> <li>□ A system to control a Air Conditioner unit</li> </ul> <p>Explain with neat diagram.</p> <p>(ii) Giving reasons for your answer suggest an appropriated control model for the Following with a neat diagram:</p> <ul style="list-style-type: none"> <li>□ An Automated Marks card Generation system</li> <li>□ An automated robot floor cleaner</li> </ul> <p>Explain with neat diagram.</p>                                                               | 10M |

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|  | <p align="center">BMS COLLEGE OF ENGINEERING, BANGALORE-19<br/>(Autonomous Institute, Affiliated to VTU)<br/><b>Department Name:</b></p> |
| <b>SECOND INTERNALS</b>                                                             |                                                                                                                                          |

|                                                    |                                   |                 |
|----------------------------------------------------|-----------------------------------|-----------------|
| CourseCode: 20CS5PCSEG                             | CourseTitle: Software Engineering |                 |
| Semester:5 <sup>th</sup> A,B,C                     | MaximumMarks:40                   | Date:01/12/2020 |
| FacultyHandlingtheCourse: Latha N.R., Sheetal V.A. |                                   |                 |
| Instructions: Internal Choice in PART C only       |                                   |                 |

### PART-A

Total 5 Marks

| No. | Question                                                       | Marks |
|-----|----------------------------------------------------------------|-------|
| 1   | Illustrate the tools included in rapid application development | 5M    |

### PART-B

Total 15 Marks

| No. | Question                                                                                                                                                                                                                                              | Marks |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2a  | Using the UML graphical notation for object classes, design Generalization hierarchy diagram identifying object classes, attributes and operations for the following :<br><br>(i) A petrol or gas station to be set up for fully automated operation. | 5M    |
| 2b  | Differentiate between the two Event-based control models                                                                                                                                                                                              | 5M    |
| 2c  | Justify how practices of extreme programming fit into principles of agile methods                                                                                                                                                                     | 5M    |

### PART- C

Total 20 marks

| No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3a  | Assume you are responsible for writing the specification for a software system that controls a network of EPOS (electronic point of sale) terminals in a store. The system accepts bar code information from a terminal, queries a product database and returns the item name and its price to the terminal for display. The system must be continually available during the store's opening hours.<br><br>(i) Model architecture for the above system with reasonable assumptions about the system requirements and justify your answer.<br>(ii) Draw Context model and Dataflow model | 10M   |
| OR  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>3b</b> | <p>(i) Draw a sequence diagram showing the interactions involved when student's registers for a course in a university. Courses may have limited enrolment, so that the registration process must include checks that places are available. Assume that the</p> <p>student accesses an electronic course catalogue to find out about various courses</p> <p>(ii) Draw a state machine model for an automatic washing machine that models different programs for different types of clothes</p> | 10M |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>4a</b> | <p>(i) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate structural model that might be used as a basis for managing the development of the following systems:</p> <ul style="list-style-type: none"> <li>□ An Aadhar Authentication system</li> <li>□ An automated Vending machine Control System</li> </ul> <p>Explain with neat diagram.</p> <p>(ii) Giving reasons for your answer suggest an appropriated control model for the Following with a neat diagram:</p> <ul style="list-style-type: none"> <li>□ A system that monitors patients in a hospital intensive care unit</li> <li>□ A system to control a refrigeration unit</li> </ul> <p>Explain with neat diagram</p> | 10M |
| OR        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
| <b>4b</b> | <p>(ii) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate structural model that might be used as a basis for managing the development of the following systems:</p> <ul style="list-style-type: none"> <li>□ An Automated radiation therapy system</li> <li>□ Digital Library system</li> </ul> <p>Explain with neat diagram.</p> <p>(ii) Giving reasons for your answer suggest an appropriated control model for the Following with a neat diagram:</p> <ul style="list-style-type: none"> <li>□ A system that monitors sugar level in patients and dispenses insulin</li> <li>□ A system to control a Television unit</li> </ul> <p>Explain with neat diagram</p>               | 10M |